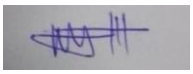
MAKERERE UNIVERSITY

PROJECT WAY-GUARD

MANAGEMENT REPORT

BY

TEAM AVRON

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2. MS. NALWOGA RACHEAL

DECLARATION

We, as Team AVRON, hereby declare that this report is our original work that has been prepared and submitted to fulfil the requirements of the Biomedical Engineering Design I course. All information, data, analyses, and insights contained herein have been independently gathered, developed, and synthesized through our collective efforts aimed at innovating a biomedical device that supports the independent mobility of visually impaired people in low-resource settings. We affirm that no portion of this work has been plagiarized, and any ideas, concepts, or materials derived from external sources have been properly cited and referenced in accordance with academic integrity standards.

Team members	**Signatures**
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Ebamu Misaki	……  ……
Nabusiita Mercy	……  ……
Tusuubira Hope Jibril	……  ……

ACKNOWLEDGEMENT

We extend our sincere gratitude to the Almighty God for granting us life, health, and wisdom to to handle this semester's work.

We also profoundly thank our mentors and instructors, **Dr. Robert Ssekitoleko** and **Ms. Nalwoga Racheal**, for their exceptional guidance, expertise, and unwavering support. Their contributions have been transformative to our progress in the Biodesign process.

ABSTRACT

This interim management report outlines the progress made by Team AVRON in the Biomedical Engineering Design I course at Makerere University. The project uses the Biodesign process to discover and address unmet medical needs through innovative engineering approaches.

Throughout the semester, the team carried out extensive research, visited healthcare facilities, and engaged with various stakeholders to identify and refine potential areas of concern. Major activities completed include needs exploration, the development of well-defined problem statements, and the generation of concepts for a solution to our final problem: the real-time detection of environmental hazards to support the independent mobility of visually impaired people in low-resource outdoor settings.

The report presents the team's structure, methodology, and collaboration approaches, highlighting essential project management practices such as risk evaluation, stakeholder involvement, and ongoing improvement. Significant challenges, including conflict management and the refinement of problem statements, were addressed methodically through structured approaches. This report demonstrates the team's commitment to promoting innovation, strengthening teamwork, and ultimately developing impactful biomedical solutions.

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LIST OF ACRONYMS

Acronym	Meaning
CDC	Centers for Disease Control and Prevention
WHO	World Health Organization
UNICEF	United Nations Children's Fund
MoH	Ministry of Health
LMICs	Low and Middle Income Countries
EVA	Earned Value Analysis
BME	Biomedical Engineering

CHAPTER 1: INTRODUCTION

1.1 Background

The Biomedical Design course in the Biomedical Engineering programme focuses on identifying, understanding, and addressing critical medical needs through innovative solutions. The course runs from the first semester of the third year until the final semester, which is the second semester of the fourth year. Besides promoting innovation, the course is designed to develop students' project management abilities. Consequently, students work in teams of individuals with different skills, abilities, and personalities, all working together to improve healthcare through innovation and technology. The course follows the Biodesign process, which guides students through several stages, including identifying unmet clinical needs, screening them to select the one the team can finally work on, and generating possible solutions through activities such as brainstorming. Each team member contributes distinct abilities in areas ranging from problem-solving and technical expertise to project management and communication, enabling the team to address real-world challenges through systematic research and collaborative design. We were assigned to a team that we named AVRON, consisting of four members, each allocated specific responsibilities within the team.

1.2 Overview of Project Management

This section presents the framework used to manage the Biodesign project, highlighting the use of organized and systematic methods to achieve the intended project outcomes.

1.2.1 Strategic Management

Strategic management involves establishing project objectives, defining success criteria, and ensuring consistency with the stages of the Biodesign process, namely Identify, Invent, and Implement. It also determines the project's scope, expected deliverables, and measurable targets.

According to Ahmed Salem Al Dhaheri (2024), strategic management entails setting a long-term vision for a project and ensuring that its activities remain consistent with broader organizational objectives and market requirements.

Our vision is to improve the independent mobility of visually impaired individuals navigating outdoor environments with limited resources, through a solution that enables real-time detection of environmental hazards. Conventional mobility aids such as white canes may not adequately detect obstacles above ground level, at head height, or on the ground. Our objective is therefore to develop an affordable and reliable assistive solution that detects obstacles at different heights and gives users timely feedback, so that they can navigate their surroundings more safely, confidently, and independently.

1.2.2 Alignment and Tool Customization

This section explains how project management tools such as Gantt charts, work breakdown structures, and scoring matrices are modified to suit the Biodesign methodology and the academic project schedule.

The project must also align with the organization's overall strategy, which focuses on either differentiation or cost leadership.

These tools are tailored to Biodesign projects because of their iterative and uncertain nature and the fixed time limits of the academic calendar. This differs from traditional project management, which generally follows a more linear and predictable process.

1.2.3 Planning and Risk Management

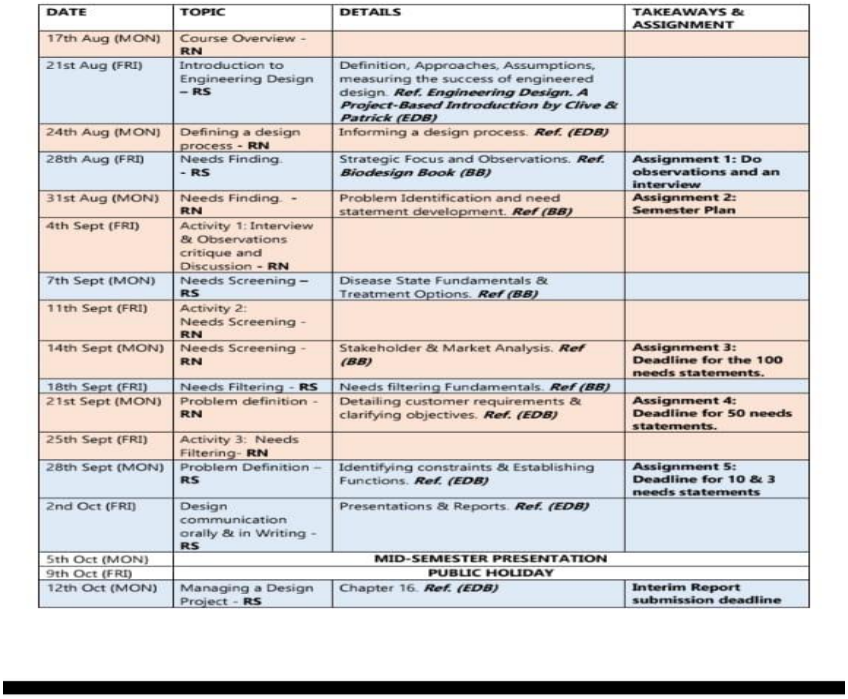
This subsection describes the project schedule, including key milestones, task dependencies, and the critical path. It also considers potential risks, such as delayed access to stakeholders and limited data, and sets out strategies to reduce or manage them ("The Role of Risk Management in Project Success: A Comprehensive Study of Project Management Practices," 2025).

The design work plan for Team AVRON

Table 1 The Team's initial design plan for needs finding

NUMBER OF WEEKS	ACTIVITY	DETAILS OF THE ACTIVITY
1	Identifying areas of focus	Choose a strategic area of focus which included maternal and child health, rehabilitation, surgery and laboratory
1	Preparation of interviews guides and observation	Define objectives; apply for hospital visits, create the questionnaire; set up tools for recording; prepare the mindset for observation, and carry out mock interview with each other in the team.
1	Ethnography and background research	Take part in conversations beyond formal interviews to gain deeper insight. Make research in the defined areas of interest.
1	Interviews	Carry out a interviews with clinicians to find out pressing clinical needs .
1	Analysis and review by BME	Present to tutors to obtain feedback and improve as advised
2	Work on improvements to problems	Make more research on the needs identified

NUMBER OF WEEKS	ACTIVITY	DETAILS OF THE ACTIVITY
3	Drafting need statements and researching each	Compose 100 need statements from the information obtained in interviews and ethnography
1	Down-selecting to the final 3 need statements	Go through needs screening based on the Biodesign process



DATE	TOPIC	DETAILS	TAKEAWAYS & ASSIGNMENT
17th Aug (MON)	Course Overview - **RN**		
21st Aug (FRI)	Introduction to Engineering Design – **RS**	Definition, Approaches, Assumptions, measuring the success of engineered design. ***Ref. Engineering Design. A Project-Based Introduction by Clive & Patrick (EDB)***	
24th Aug (MON)	Defining a design process - **RN**	Informing a design process. ***Ref. (EDB)***	
28th Aug (FRI)	Needs Finding. - **RS**	Strategic Focus and Observations. ***Ref. Biodesign Book (BB)***	**Assignment 1: Do observations and an interview**
31st Aug (MON)	Needs Finding. - **RN**	Problem Identification and need statement development. ***Ref (BB)***	**Assignment 2: Semester Plan**
4th Sept (FRI)	Activity 1: Interview & Observations critique and Discussion – **RN**		
7th Sept (MON)	Needs Screening – **RS**	Disease State Fundamentals & Treatment Options. ***Ref (BB)***	
11th Sept (FRI)	Activity 2: Needs Screening - **RN**		
14th Sept (MON)	Needs Screening - **RN**	Stakeholder & Market Analysis. ***Ref (BB)***	**Assignment 3: Deadline for the 100 needs statements.**
18th Sept (FRI)	Needs Filtering - **RS**	Needs filtering Fundamentals. ***Ref (BB)***	
21st Sept (MON)	Problem definition - **RN**	Detailing customer requirements & clarifying objectives. ***Ref. (EDB)***	**Assignment 4: Deadline for 50 needs statements.**
25th Sept (FRI)	Activity 3: Needs Filtering- **RN**		
28th Sept (MON)	Problem Definition – **RS**	Identifying constraints & Establishing Functions. ***Ref. (EDB)***	**Assignment 5: Deadline for 10 & 3 needs statements**
2nd Oct (FRI)	Design communication orally & in Writing - **RS**	Presentations & Reports. ***Ref. (EDB)***	
5th Oct (MON)	**MID-SEMESTER PRESENTATION**		
9th Oct (FRI)	**PUBLIC HOLIDAY**		
12th Oct (MON)	Managing a Design Project - **RS**	Chapter 16. ***Ref. (EDB)***	**Interim Report submission deadline**
16th Oct (FRI)	Concept Generation - **RN**	Ideation & Brainstorming - Activity format. ***Ref (BB)***	
19th Oct (MON)	Generating Design Alternatives - **RN**	Conceptual design. ***Ref. (EDB)***	
23rd Oct (FRI)	Team activity for generating design alternatives - **RN**		
26th Oct (MON)	Concept Screening - **RS**	Concept Screening Fundamentals Concept Mapping. ***Ref (BB)***	
30th Oct (FRI)	Evaluating Design alternatives and choosing a design – **RN**	Applying Metrics to objectives. ***Ref. (EDB)***	
2nd Nov (MON)	Concept selection - **RN**	Concept Selection fundamentals. ***Ref (BB)***	
6th Nov (FRI)	Working session - **RS & RN**		
9th Nov (MON)	Team progress check - **RS**		
13th Nov (FRI)	Working session – **RS & RN**		
25th Nov 2026	**FINAL PRESENTATION**		
4th Dec 2026	**LOGBOOK & FINAL REPORT SUBMISSION**		

Figure 1 The group schedule (Gantt chart)

Needs finding was to be carried out through interviews, observations and in-depth research in the first four weeks of the semester. This was not done on schedule, which pushed back the entire programme. Because time was short, we did not secure a clinical client early enough. This later left us with too little information about our need, and we had to correct our problem statements. Once these were corrected, we moved forward well.

1.2.4 Quality Management and Continuous Improvement

This subsection sets quality standards for data collection, analysis, and documentation, and defines validation checkpoints and iterative feedback loops to refine the methodology.

Project quality management ensures that a project meets or exceeds customer requirements through a structured approach of quality planning, quality assurance, and quality control. It involves defining quality standards, putting processes in place to meet them, and monitoring results to prevent defects and ensure customer satisfaction (Shah et al., 2022).

We met our lecturers at every milestone to confirm that we were on the right track. These meetings helped us correct our information, and the lecturers consistently urged us to research everything we presented in depth.

1.2.5 Communication and Stakeholder Engagement

Stakeholder engagement in project management is the process of identifying, communicating with, and managing relationships with individuals or groups who have an interest in or influence over a project. These may include clients, sponsors, team members, regulatory bodies and end users (Nauman et al., 2025).

For our project on mobility for visually impaired people, we engaged stakeholders including a biomedical Engineer at Ruby hospital, Eng. Allan Rodin Gidoi, after which we made further research.

1.2.6 Cost Control and Earned Value Analysis

Resource allocation and finances were tracked closely throughout the project to manage time, budget, and materials with discipline. An Earned Value Analysis (EVA) framework was set up as the main quantitative tool for assessing project performance. It allowed the team to compare planned budgets with actual spending while tracking progress against deliverables and milestones. This integrated approach gave objective measures for spotting schedule and budget variances early, so that corrective action could be taken in good time.

All resource investments were recorded systematically. Spending was grouped into personnel time (hours spent on needs finding, clinical consultations, concept development, screening iterations, and documentation), material costs (literature, office supplies, prototyping materials), and use of facilities (access to clinical sites for observations and interviews). Personnel hours were allocated by competency and task requirements: specialized activities such as statistical analysis, and medical device literature synthesis needed concentrated time from members with, while collaborative activities such as brainstorming, meetings, and consensus-building needed the full team (What Is Cost Control in Project Management?, 2025).

1.3 Project Objectives

This project aims to systematically identify unmet healthcare needs through structured clinical research, develop evidence-based problem statements, and apply the Biodesign methodology to generate and evaluate innovative engineering solutions for the real-time detection of environmental hazards by visually impaired people navigating low-resource outdoor environments. Specific objectives include carrying out ethnographic needs assessment at specialized healthcare facilities to recognize genuine clinical gaps and translate observations into solution-neutral problem definitions, and generating diverse concepts through structured brainstorming and narrowing them with weighted screening criteria that prioritize clinical impact, technical feasibility, and suitability for low-resource settings.

The project further aims to build professional skills in technical report writing that demonstrates sound engineering analysis, management report writing that reflects disciplined project governance and resource stewardship, and persuasive presentation, so that students can explain complex biomedical innovations to clinicians, administrators, investors, and policymakers within limited time. Through individual logbooks recording design decisions, iterative learning, and reflection, students will develop the technical, communication, and leadership competencies that biomedical engineers need to contribute to global healthcare innovation. Ultimately, the project seeks to produce deliverables with enough technical rigour and clinical relevance to inform later prototype development and clinical validation, while shaping engineers who understand that technology alone is not enough without genuine stakeholder engagement, practical design trade-offs, and sustained commitment to turning concepts into deployed solutions that serve vulnerable populations in resource-constrained global health contexts.

1.4 Project Plan

We designed an approach that lets us manage risk by anticipating likely challenges. The project work plan has been essential in giving us a focused roadmap for carrying out and completing the project.

Table 2 The team's project plan

Group activity	Outcome
Assigning roles to members	Each team member is assigned a role to carry out
Needs finding	Using different techniques, 100 needs are generated
Needs screening	Using a defined set of criteria (a MAUT model, then a C1–C4 feasibility matrix), three needs are selected from the many
Problem statements	A problem statement is written for each of the three top needs

Group activity	Outcome
Selecting the one need to work on	From the three problem statements, one need, the real-time detection of environmental hazards for visually impaired people, is selected as the project
Concept generation	Various concepts are produced for solving the one chosen need
Concept selection	The best concept is chosen to work on

1.5 Project Timeline

This subsection presents a chronological schedule of key activities and milestones, often shown as a Gantt chart or timeline, to track progress and keep to the deadlines.

Table 3 The team's project timeline

Task	Start date	End date	Period (days)
Introduction and Finding a team name	Mon 17th Aug	Fri 21st Aug	5 days
Identifying focus areas	Mon 24th Aug	Fri 28th Aug	5 days
Drafting recommendation letters to health facilities and drafting questionnaires	Mon 31st Aug	Fri 4th Sept	5 days
Background research	Mon 7th Sept	Fri 11th Sept	5 days
Exploring needs at the selected hospitals through interviews and observations	Mon 14th Sept	Fri 18th Sept	5 days
Developing need statements	Mon 21st Sept	Fri 25th Sept	5 days
Needs screening	Mon 28th Sept	Fri 2nd Oct	5 days
Problem statements	Mon 12th Oct	Fri 16th Oct	5 days
Concept generation and Mid semeter reports	Mon 19th Oct	Fri 23th Oct	5 days
Concept selection	Mon 26th Oct	Fri 30th Oct	5 days

Task	Start date	End date	Period (days)
Preparation of final reports	Mon 2nd Nov	Fri 6th Nov	5 days
Down-selecting to the final concept and preparing for the final presentation	Mon 9 th Nov	Fri 13th Nov	5 days

1.6 Project Strategies

To carry the project through successfully, we keep up regular contact with our course instructors to get expert advice, clear up doubts, and receive constructive feedback. Their guidance has given us a better grasp of the course content and of the specific problem we are solving. We also research extensively in academic databases and scientific publications to understand the underlying issues, existing solutions and gaps, so that our work stays evidence-based and relevant. Combining thorough investigation with mentorship equips us to approach the problem with clarity and innovation.

Within the team, we stress organization and accountability to keep collaboration smooth. We hold regular meetings to discuss progress, resolve issues, and assign specific tasks. Roles are clearly defined so that everyone contributes effectively and uses their strengths. Time management is another key part of our strategy: we prioritize meeting deadlines and keeping to schedules. Communication is central, with virtual platforms such as WhatsApp and e-mail combined with in-person interaction to keep all members informed.

1.7 Team Policies and Finances

This section outlines the rules that govern the team, such as decision-making protocols, attendance policies and financial management, including budget allocation and funding sources.

Team policies. As a group, we agreed on policies for managing the project well: working within the fixed time we have, keeping discipline among members, and sharing work equally. They are summarized below.

Table 4 The team policies

Description	Fine
Coming late to meetings	The individual is given an extra activity and told to complete it within a set period
Missing group activities without prior communication	The individual pays a fine of UGX 5,000
Submitting assigned work after the deadline	The individual pays a fine of UGX 2,000

Finances. To keep the project running smoothly, each group member contributes UGX 5,000 per month to our treasurer. These contributions form a shared pool that supports various project needs, so that financial constraints do not stop us from progressing. This promotes accountability and a sense of shared responsibility and commitment to the project's success. Managing the contributions also builds our financial skills, such as budgeting and expense tracking.

CHAPTER 2: TEAM ORGANIZATION

This chapter describes how the team is structured and how it operates to achieve the project goals, with emphasis on collaboration and clarity of roles.

2.1 Roles and Their Responsibilities

This section details the roles within the team and their specific duties, to ensure accountability and efficient distribution of tasks.

Table 5 The team's roles and responsibilities

Name	Position	E-mail
Nassanga Angel Ruth	Team Lead	angelruthnassanga@gmail.com
Ebamu Misaki	Research Lead	ebamumisaki@gmail.com
Nabusiita Mercy	Secretary	nabusiitam@gmail.com
Tusuubira Hope Jibril	Finance manager	thopejibril@gmail.com

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2.1.1 Team Lead

The Team Leader's responsibilities include overseeing project progress, making strategic decisions, and resolving conflicts.

- Makes sure the group sets the vision, aim and mission of the project.
- Oversees progress against the agreed timeline and assigns tasks.
- Leads the group in making key project decisions and settling conflicts.
- Encourages collaboration among members and ensures effective communication.

2.1.2 Research Lead

- Leads the research process by identifying relevant information, sources, and evidence related to the project.
- Conducts literature reviews and gathers data to understand the problem, existing solutions, and knowledge gaps.
- Guides the team in developing research questions, identifying clinical needs, and formulating clear problem statements.
- Ensures that research findings are accurate, reliable, evidence-based, and properly documented.
- Coordinates research activities among team members and supports the synthesis and analysis of collected information.

- Prepares research summaries, reports, and presentations to communicate findings and inform team decisions.

2.1.3 Secretary

The Secretary documents meetings, maintains records, and ensures that decisions and action items are communicated clearly.

- Schedules meetings, prepares agendas, and records detailed minutes.
- Keeps track of project deadlines so the group stays organized and on schedule.
- Manages and organizes project documentation so that files are accessible to the team.
- Supports communication within the team and with external stakeholders, ensuring timely and accurate information flow.

2.1.4 Finance Manager

The Finance Manager handles budgeting, tracks expenses, and reports on the financial status to ensure fiscal responsibility.

- Develops and manages the project budget, ensuring funds are allocated appropriately.
- Tracks expenses, keeps financial records, and gives regular updates on the budget.
- Advises the team on financial feasibility when considering project options or design changes.
- Ensures that project spending follows the project's financial plan.
- Keeps financial reports so that stakeholders are informed about the financial status.

2.2 Mission

The team discussed what a worthy mission statement could be, based on our shared desires, strengths and passion. The statement symbolizes what we collectively stand for and what we hope to achieve.

Mission statement: To design biomedical technologies that are accessible across all healthcare environments, regardless of resources or infrastructure, with the goal of saving and improving lives, particularly by enabling independent mobility for people living with visual impairment.

2.3 Core Values

These values let the group reflect its mission and vision and drive our commitment to creating innovative, locally produced medical devices that are affordable and widely accessible. They are the guiding principles behind our decisions, our collaboration, and our strategy for achieving our objectives.

- **Teamwork:** Highlights the importance of collaboration in improving healthcare accessibility and delivery. By drawing on the diverse skills, perspectives and strengths of our members, we foster innovation and create effective, well-rounded solutions to the problem of mobility for visually impaired people.

- **Time management:** Helps us meet deadlines according to the schedules we set. Some of our deadlines were not met on time, but we kept pursuing them and eventually achieved them. We also schedule meetings as a group and with the people we work with on our project, such as our tutors, and we manage the time.
- **Flexibility:** Reflects our adaptability to changing circumstances and emerging needs. It keeps the team open to new ideas and constructive feedback on our need statement, problem statement and other evolving project requirements.

2.4 Team Focus Areas

The team discussed strategic areas of interest and focus. The pool included:

1. Maternal and child health
2. Emergency medicine
3. Rehabilitation medicine (the area that underpins our topmost need, independent mobility for visually impaired people)

CHAPTER 3: TEAM ACTIVITIES

3.1 Places Visited and Interviews Conducted

The first place we visited for needs finding was Kawempe National Referral Hospital, where we interviewed Ms. Sauda Nakalazi, a midwife and labour ward in-charge, about maternal and neonatal challenges they face. Next, we had an interview with Eng. Allan Rodin Gidoi a Biomedical Engineer at Ruby hospital, about the biomedical and technical challenges faced in the hospital.. Finally, at Mulago Molecular labs, we interviewed Mr. Laymond Kalema a laboratory technologist under immunology department , which further informed our analysis as we developed a definitive needs statement. These observations and interviews followed the Biodesign methodology, which requires the systematic identification of people, places, processes, and products as the framework for developing effective need statements.

Table 6 The Clinical liaisons for our project

Name	Position held	Contact
Eng. Allan Rodin Gidoi	Biomedical Engineer at Ruby Hospital Kampala	0751619717
Ms. Sauda Nakalazi	Midwife and labour ward in-charge ata Kawempe National Referral Hospital	0704544493
Mr. Laymond Kalema	A Laboratory Technologist at Mulago Molecular biology labs	0706663966
Eng. Nicholas Mbowa	Biomedical Engineer at Kawempe National Referral Hospital	0703964920

3.2 Research

Our research ran through every stage, from needs exploration and need statement development to needs screening and needs selection, drawing on a variety of authoritative sources to gather solid information on the unmet needs we identified. We searched major academic databases, specifically PubMed and Google Scholar, for scholarly articles and studies. At the same time, we consulted well-known and trusted official health organization websites, including the Centers for Disease Control and Prevention (CDC), the World Health Organization (WHO), the United Nations Children's Fund (UNICEF), and the local Ministry of Health (MoH), so that our needs analysis was backed by both peer-reviewed evidence and official public health guidelines and data. For our topmost need, the WHO's global estimates on vision impairment were central to our evidence base.

3.3 Meetings

In the due course of the semester the team has kept steady progress through structured communication. We met at least weekly with a course instructor, after deciding as a team which lecturer could best assist, and the secretary handled scheduling by e-mail and calendar invitations. In these sessions we discussed progress and challenges and decided the way forward. Besides these instructional meetings, the team held regular internal discussions to track progress, delegate responsibilities, and plan next steps. The secretary recorded the minutes of all meetings and shared them with the team and the relevant instructors on the team's Google Drive for full documentation.

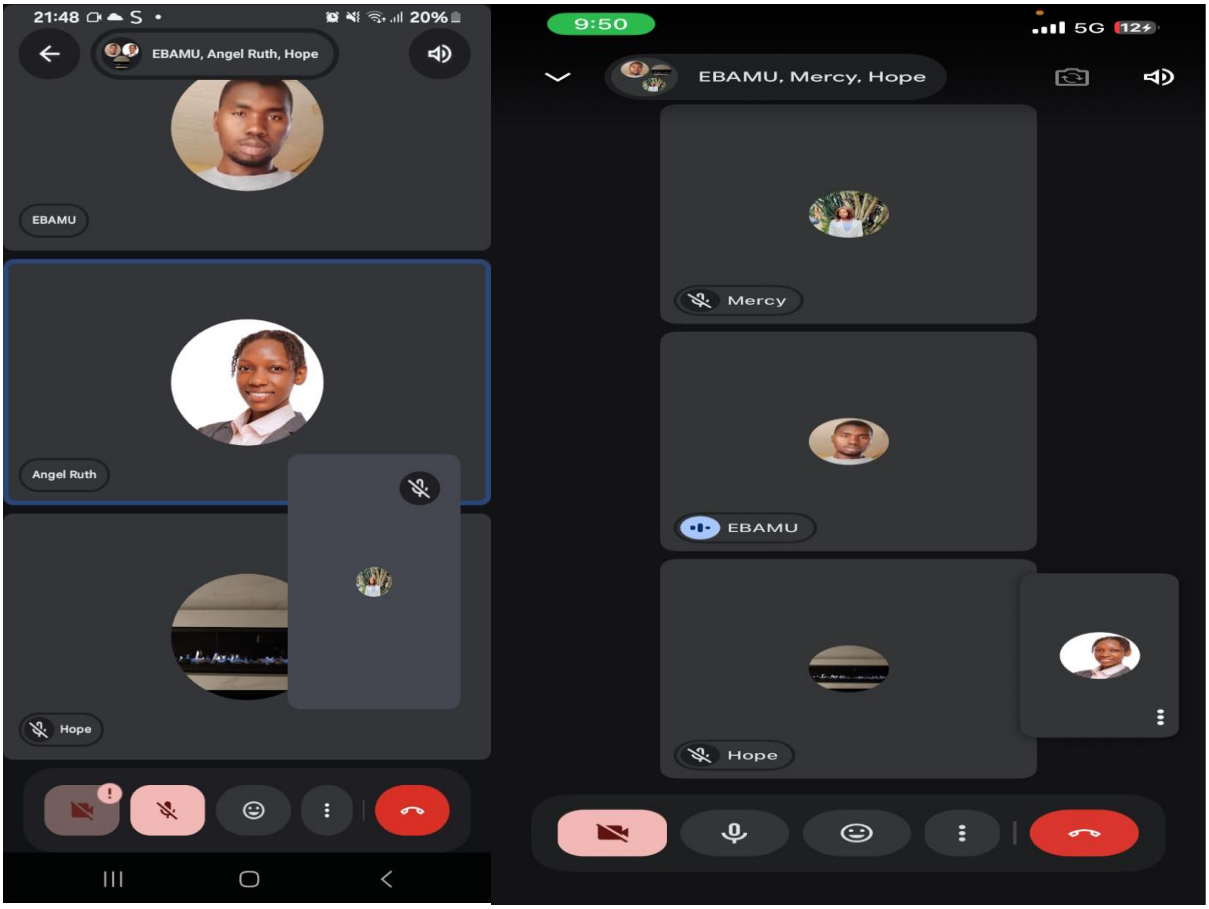


Figure 2 Team meetings held online

3.4 Team Building

Effective team building is crucial because it supports collaboration, sharpens communication, and raises overall team productivity. Research confirms that structured team-building activities foster higher levels of trust and cooperation among members and thereby improve team performance. Interventions such as workshops, team outings, and problem-solving exercises help strengthen relationships and cooperation. Team building that deliberately builds on individual strengths while

promoting teamwork also produces more cohesive groups that adapt better to fast-changing work environments ((PDF) Does Team Building Work?, n.d.). In addition to lifting morale, team-building initiatives support conflict resolution and engagement. Teams with strong interpersonal cohesion tend to resolve conflicts more constructively because they understand each other's working styles and perspectives. Strong teams also encourage innovation through open discussion and diverse ideas, which improves decision-making and problem-solving (Edmondson & Lei, 2014).

3.5 Communications

As a team, we relied mainly on WhatsApp for internal communication, finding it the easiest way to talk virtually among members (Ujakpa et al., 2018). It was our main channel because it was fast, informal, and highly accessible, which let us share project updates, coordinate meeting logistics, send timely reminders, and exchange ideas in real time. The group chat format kept everyone engaged and informed, with discussion flowing during and outside planned meetings. For communicating with lecturers, coordinators, and certain stakeholders we used e-mail, which was particularly helpful for scheduling meetings. Phone calls were used for urgent issues that needed immediate attention.

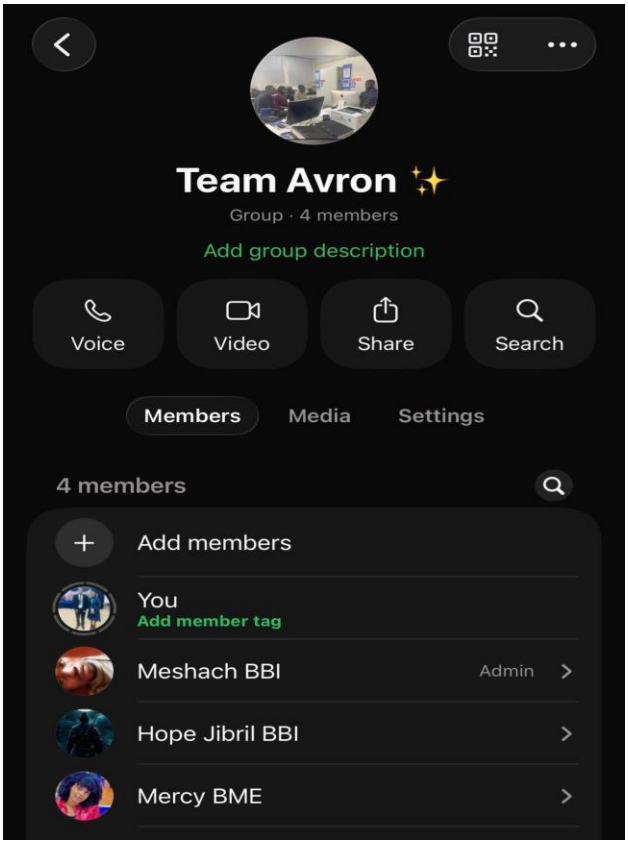


Figure 3 Team WhatsApp group platform

3.6 Needs Finding

To find unmet healthcare needs, the team combined literature reviews, clinical site visits, and interviews in a systematic way. Early searches on PubMed and Google Scholar built a knowledge base, shaped our interview questions, and gave us a way to validate field findings. Over two weeks we visited facilities including Kawempe National Referral Hospital, Ruby Hospital Kampala and Molecular labs Mulago to study maternal and child health, laboratory work, and rehabilitation. Using the 4Ps framework (Process, Place, People, and Product), our observations revealed key issues such as slow service delivery, problems with medical waste handling, challenges in managing bedridden patients, and the lack of affordable assistive devices for people with disabilities. Semi-structured interviews with professionals added depth to these observations. Despite slight information gaps caused by ethical constraints, the process allowed us to generate and validate 100 need statements, which were refined with instructor guidance before needs screening.



Figure 4 Interview with Mr. Laymond Kalema a Laboratory Technologist at Mulago Molecular Biology Labs

3.7 Needs Screening

After needs finding, the team had 100 potential healthcare needs and, guided by the Biodesign book, began a multi-stage filtering process. The pool of 100 was narrowed to 70, then to 50, then to 20 , then to a top 10 using a Multi-Attribute Utility Theory (MAUT) model that scored each need on market size, urgency, provider workflow, engineering feasibility, safety, and suitability for low-resource settings. The top 10 were then passed through a four-criterion operational feasibility matrix (C1 clinical and stakeholder integration, C2 technical and prototyping scope, C3 regulatory risk and safety profile, C4 cost of goods and consumable burden). Our final top three needs were presented to the class and instructors, with the real-time detection of environmental hazards for visually impaired people as the topmost need. The presentations let the team show its understanding of needs screening and gain peer and instructor feedback (Yock et al., 2015). Full tables are in the technical report.

CHAPTER 4: CHALLENGES AND CONFLICT MANAGEMENT

This chapter outlines the practical steps and strategies the team used to manage the project from needs finding right up to concept screening and presentation. It covers how the team handled key operational areas: implementing and enforcing policies on member conduct, resolving conflicts effectively to keep collaboration smooth, managing finances to allocate resources efficiently, and assessing how far tasks were accomplished. It also records the specific challenges met and ends with the conclusions the team drew from the whole experience.

4.1 Introduction to Conflict Management

Conflict is a normal part of teamwork and often begins with simple disagreements about ideas or how roles are carried out. Peace and harmony are desirable, but conflict can push a team to explore different and better ways of working. When managed well, it can yield increased productivity and innovative solutions, ultimately leading to win-win results. Poorly managed conflict can lead to team dysfunction and failure. Any team striving to meet its objectives must accept that conflict is inevitable in a work environment. It often stems from differing worldviews, belief systems, personal egos, perspectives, and backgrounds.

The main causes of disputes among team members over the semester were:

- **Poor time management:** This was a major problem, especially for group meetings. Even though we sent notices of the date and time, members often struggled to arrive on time (Chin & Hamid, 2015). There was also confusion about the meeting location and whether a meeting would be held in person. When asked about lateness, members usually said they were too busy or had forgotten. Several members also often neglected assigned tasks. This lack of initiative spread the workload unevenly, so self-driven members ended up doing most of the work. Some members felt overburdened while others stayed disconnected from the project. These imbalances worsened existing tensions and hurt the team's productivity and ability to work together.
- **Fear of taking up some roles and responsibilities:** Some members did not want to be given roles, which caused a serious argument before they were persuaded to accept them.
- **Not completing assigned tasks on time:** Some members repeatedly missed the deadlines set by the chairperson, which delayed the team's progress.

Despite these obstacles during needs finding, needs screening, concept generation and selection, the team handled them through specific skills and a structured resolution process, which settled the disagreements and strengthened our unity and collaboration, as explained below.

4.2 Strategies for Conflict Management

Open discussion. The chairperson consistently promoted open and honest discussion, giving members a platform to raise concerns with one another. In team meetings, the chairperson encouraged members to share their thoughts about those who had not contributed. This open

dialogue motivated members to improve their efforts, which fostered collaboration and supported the project's success.

Collaborating. This means working with all parties to find a mutually acceptable solution. For example, the chairperson could assign tasks to each member, but some did not meet expectations, producing weak presentations that drew many comments from the course mentors and often sent the whole team back to revise. This frustrated members who had done their part. To resolve this, the chairperson encouraged the whole team to work together, ensuring high-quality, well-researched work.

Focusing on resolution instead of blame. In internal meetings, conflict discussions centred on finding solutions rather than assigning blame, because blaming individuals could spark arguments and waste valuable time.

Introducing and implementing fines. To address poor time management, the team agreed that any member arriving late to a meeting without prior notice would pay a fine, set at the time of arrival, to the treasury.

4.3 Key Steps in Conflict Resolution

The conflict management process involved these steps:

- **Identifying the conflict:** We acknowledged that a dispute or state of disharmony existed and was hurting the team's peace and productivity. This was often triggered by visible signs such as consistent lateness, failure to submit assigned work, or open frustration in meetings (Behfar et al., 2008).
- **Conflict analysis:** We investigated the nature, scope, and impact of the dispute, assessing its severity and which members or project areas were most affected. This gave a clear, objective picture needed before generating and selecting solutions (Min et al., 2018).
- **Alternative solution identification:** The team brainstormed a wide range of options, not just the obvious ones, including changes to roles, communication, or new policies, so that we could carefully select the single best solution that was effective, fair, and acceptable to everyone.
- **Applying conflict resolution techniques:** The team put the agreed solutions into practice. For example, fines for lateness were actually introduced and collected, and for collaboration the chairperson directed the whole team to work together on the task that had previously failed.
- **Choosing the best alternative:** The team evaluated the options and compared the pros and cons of each against fairness and sustainability standards, then made an informed decision to adopt the alternative with the highest likelihood of success.
- **Implementing the solution in line with the team's governing rules:** The team carried out the chosen plan while following the policies, principles, and procedures it had previously agreed on.

With this systematic approach, we handled disputes, rebuilt team unity, and sharpened our collaborative skills. It eased the pressing challenges and gave us tools for sustained teamwork and professional work (Greer & Dannals, 2017).

Figure 4: The team's conflict-resolution process – (1) identify conflicts in the project; (2) analyse the conflict; (3) identify alternative solutions; (4) apply conflict resolution techniques; (5) choose the best alternative; (6) implement the solution to the conflict; (7) review the impact, and return to step 1 if needed.

4.4 Challenges Faced

Throughout the project, our team met several obstacles that affected our progress, collaboration, and execution. They arose from personal, logistical, financial, and academic factors, but we adapted strategies to reduce their effects where possible. The key issues were:

1. **Unfamiliarity with the design process:** At the start of the semester, the unusual nature of the course, which differs from other units, made its requirements hard to grasp. This led to initial complexity, time-consuming revisions of assignments, and overall time loss. With instructor guidance and experience, we began to master the expectations, improve our time management, and adapt to the ongoing refinement process, which improved our understanding of design principles and project management.
2. **Financial constraints for resources:** We had a shortage of funds for transportation to the hospitals, printing some materials, caused by economic hardship affecting all members, which limited our ability to produce high-quality deliverables on time.
3. **Poor communication among members:** Inadequate communication recurred, with some members failing to tell others about personal problems or conflicts. This contributed to missed meetings and a lack of alignment in the group.
4. **Conflicting academic schedules:** The fixed university timetable created barriers, particularly for members repeating course units from earlier years. These overlaps prevented full attendance at some sessions and complicated group coordination.
5. **Missed meetings due to unforeseen events:** Some members could not attend scheduled meetings because of unexpected circumstances such as illness. This disrupted our workflow and required rescheduling, which occasionally delayed decisions.
6. **Delays in obtaining hospital permissions for clinical observations:** Approvals from hospitals often arrived well after our deadlines, leaving us unable to proceed on several occasions, and even when granted we could not always hold interviews with clinical clients as scheduled. To cope, we used a focused strategy during visits: concentrating on topics covered in class and asking direct, targeted questions. This reduced confusion and made the most of our limited time with hospital staff, so observations stayed productive despite the setbacks.

These challenges tested our resilience but built teamwork, adaptability, and problem-solving. By addressing them proactively, we kept our momentum and continued toward our goals.

CONCLUSION

Team AVRON's Management Report summarizes our journey through Biomedical Engineering Design I, showing how diligent research, collaboration, and a structured approach to problem-solving enabled us to identify critical unmet needs and gaps in existing healthcare solutions. Our topmost need, the real-time detection of environmental hazards to support the independent mobility of visually impaired people, emerged from this process. The insights gained from meetings with experts and tutors were essential in refining our project focus and clarifying our objectives, and the team stressed the importance of precise language and effective project management throughout. The financial management strategies we put in place kept the project running smoothly and built a strong sense of accountability and shared responsibility, equipping the team with vital skills in budgeting and resource prioritization. Going forward, the team remains committed to iterating on its findings, exploring innovative solutions, and drawing on the collaborative spirit shown here to make a meaningful and positive difference to the lives of visually impaired people and to patient care and outcomes in the future.

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APPENDICES

Appendix: Minutes of AVRON meetings

Team Avron Meeting Minutes

Clinical Client Interview with a molecular laboratory staff.

Date	9 September 2026
Time	4:00 – 5:00 pm
Interviewee	Raymond Kalema, Molecular Laboratory
	Mercy Nabusiita, Angel Ruth Nassanga, Hope Jibril Tusubila, Misaki Ebamu
Purpose	To know and understand how a laboratory operates, a challenges and risks faced and areas for improvement.

1. Sample Handling

- Samples are received from different parts of the country.
- Sample types include DNA and RNA.
- Some samples are processed immediately, while others are stored for later.
- Work is segregated to prevent contamination.

2. Key Challenges

- Sequencing is the most difficult part of the work.
- Unstable power supply.

3. Risks

- Accidental infections.

4. Causes of Errors

- Incompetency, including staff using a machine for a purpose it is not designed for.

5. Equipment Breakdown

- The lab has a backup lab with the same machines, which is used if equipment fails.

6. Areas for Improvement

- Speeding up work and shortening turnaround time.

Action Items

- Develop needs from the information gathered from Mr. Kalema during the interview.

Minutes prepared by: Mercy Nabusiita (Team secretary)

Team Avron Meeting Minutes

Interview with a Biomedical Engineer

Date	9th September, 2026
Time	1:00pm to 2:00pm
Mode	Online meeting
Interviewee	Allan Rodin Gidoi, Biomedical Engineer, Ruby Hospital
Attendees	Allan Rodin Gidoi, Mercy Nabusiita, Angel Ruth Nassanga, Hope Jibril Tusubila, Misaki Ebamu
Purpose	To understand a typical day in the work of a biomedical engineer, along with the challenges, priorities and opportunities for innovation in the field.

1. A Typical Day

- Carries out facility checkups, moving between hospital departments to identify equipment that is due for service.
- Plans for surgeries.
- Checks the hospital manifold that supplies oxygen to the hospital.

2. Challenges

- Technical and regulatory challenges.
- Poor communication among the different professions, since biomedical engineering is multidisciplinary.
- Lack of calibration tools for some devices.

- Expensive spare parts. Some are not made locally, so sourcing them from abroad is costly.
- Poor procurement can lead to a lack of spare parts at a facility.

3. Priorities When Resources Are Limited

- Priorities depend on the needs of the facility, which vary from one facility to another. A priority at one facility may not be a priority at another.

4. Areas Needing More Innovation

- Maternal health mostly because it has a massive impact.

Action Items

• Develop needs from the information gathered from Mr Allan Rodin Gidoi during the interview.

Minutes prepared by: Mercy Nabusiita

Team Avron Meeting Minutes

Consultation with Madam Racheal Nalwoga on our Need Statements

Date	15 September 2026
Time	3:00 – 3:30 pm (30 minutes)
Facilitator	Madam Racheal Nalwoga
Attendees	Madam Rachel Nalwoga, Mercy Nausiita, Angel Ruth Nassanga, Misaki Ebamu Jibril Hope Tusubila
Purpose	To discuss the need statements the team had developed and get the facilitator's feedback.

1. Wording of Need Statements

- Avoid subjective or loaded words such as "safe", "accurate" and "invasive".

2. Scope and Length

- Some need statements were too broad. They should be brief and precise.
- Need statements should not include explanations.

3. Structure of a Need Statement

- Each need statement should contain only three elements: the population, the problem and the outcome.
- Each need statement should have only one outcome. For example, treatment and detection cannot appear in the same statement.

4. Screening of Needs

- The need statements should be improved before moving on to screening.
- The team decides the criteria (filters) used to screen the needs.

Action Items

- Revise the need statements in line with the feedback from Madam Racheal
- Agree on the criteria for screening the needs.

Minutes prepared by: Mercy Nabusiita

Group Meeting Minutes

Date: Monday, 28 September 2026

Purpose: Design Group Work planning

Attendees: Misaki Ebamu, Hope Jibril Tusubila, Mercy Nabusiita, Angel Ruth Nasanga

Agenda and Proceedings

1. Opening prayer
2. Division of work
3. Submission deadline
4. Closing prayer

Work Allocation

Task	**Assigned to**

Need 1	Misaki Ebamu
Need 2	Hope Jibril Tusubila
Need 3	Mercy Nabusiita
Presentation slides (compiled from the findings on the three needs)	Angel Ruth Nasanga

Deadline

All members are to submit their assigned work by Wednesday evening, 30 September 2026.

Closing

The meeting ended with a closing prayer.

Date: Tuesday, October 13, 2026

Time: 8:00 pm to 9:00 pm (1 hour)

Attendees: Mercy Nabusiita, Angel Ruth Nassanga, Hope Jibril Tusubira, Misaki Ebamu

Purpose: Plan the mid-semester presentation on Friday, October 16, 2026.

1. Presentation Content

The presentation will cover progress made so far this semester:

- **Needs finding:** Hospital visits, interviews with clinical clients, background research, and the resulting list of 100 needs.
- **Needs screening:** Narrowing from 100 needs to 70, 50, 20 then to 10 and finally to the top 3.
- **Problem statements:** Generated from the top 3 needs (presented last week).

2. Task Assignments

Section	**Assigned To**
Needs finding	Mercy and Angel
Needs screening	Jibril and Misaki
Problem statements	To be finalized at the next meeting

3. Mid-Semester Report

The group briefly discussed the report. Sections the report must include still need to be confirmed, and this will be looked into at the next meeting.

4. Action Items

- **Mercy and Angel:** Prepare the needs finding section.
- **Jibril and Misaki:** Prepare the needs screening section.
- **All members:** Review the problem statements and report requirements before the next meeting.

5. Next Meeting

Date: Wednesday, October 14, 2026

Venue: In person, in a discussion room at the Main Library

Agenda: Problem statements section and mid-semester report structure

Minutes prepared by: Mercy Nabusiita

Team Avron Meeting Minutes

Date: Thursday, October 15,2026

Time: 8:00pm to 9:00pm

Venue: Online (the planned physical meeting at the Main Library on Wednesday did not take place)

Attendees: Mercy Nabusiita, Angel Ruth Nassanga, Hope Jibril Tusubila, Misaki Ebamu

Purpose: Check each member's progress, combine the presentation, and assign presenters for Friday's mid-semester presentation

1. Progress Check

Each member reported on the sections assigned at the previous meeting:

Needs finding: Mercy and Angel

Needs screening: Jibril and Misaki

All members had completed their parts.

2. Combining the Presentation

During the meeting, the group combined all completed sections into one PowerPoint presentation.

3. Presenter Assignments

No.	Section	Presenter
1	Introduction and areas of focus	Mercy
2	Needs finding process to developing needs statements	Angel
3	Needs screening	Misaki
4	Top three needs and conclusion	Jibril

4. Action Items

All presenters: Prepare to present their assigned sections in the combined PowerPoint on Friday.

Minutes prepared by: Mercy Nabusiita (Team secretary)